1. A road network has parallel roads, which are equidistant from each other and running North-South or East-West only. The road junctions A, B, C, H and X are such that A is East of B and West of C, H is South-west of C and South-East of B. B is South-East of X. Which of the junction are the farthest South. And the farthest East?
   (a) H, B (b)H,C
   (c) C, H (d)B, H
2. Four players A, B, C and D have to form into two pairs, however, no pair can play together more than seven times in a row A and B have played seven games in a row. C and D have three in a row. C does not to work with A. Who should play with B?
   (a)A (b)D
   (c)C (d)Cannot be determined
3. **If ROSE** is coded as 6821, CHAIR is coded as 73456 and **PREACH** is coded as 961473, then the code for **SEARCH** is
   (a)216473 (b)214673
   (c)214763 (d)246173

Direction[Q. Nos. 4 to 6] Cricket club in five towns A, B, C, D and E have one team each named P, Q, R, S and T, not necessarily in the same order.
The team in A has beaten R, P and S, Q has beaten the teams in E, C and A. Team R is in B and the team in C is not S.

4. Where is the team Q?
   (a) A (b)B (c)C (d)D
5. Where is the team P?
   (a) A (b)B (c)C (d)D
6. Which team is in A?
   (a)P (b)Q (c)S (d)T
7. Find the number that comes that comes next in the series 120, 99, 80, 63, 48,....
   (a)35 (b)38 (c)39 (d)40
8. In a certain school, the number of student in each section was 24. After admitting some students,three new sections have been started and now there are 16 sections with 21 students in each. What is the number of newly admitted students ?
   (a)14 (b)24 (c)16 (d)26
9. The nine alphabets L, M, N, O, P, Q, R, S and T are assigned to nine integers 1 to 9 not necessarily in the same order 4 is assigned to P. The difference between P and T is 5. The difference between N and T is 3. What is the integer assigned to N?
   (a)7 (b)6 (c)5 (d)4

Direction [Q. Nos. 10 to 13] : Five boys A, B, C, D, E, and five girls P, Q, R, S, T are standing in two rows facing each other not necessarily in the order. E is not at any ends. C is to the immediate right of B and D is to the immediate left of A, who is facing P. There are as many girls between P and Q as between R and S. A is second to the left of B.S and R are not facing either B or D.

10. Which pair of boys are standing at the ends of the the row ?
    (a)C and D (b)C and B
    (c)D and B (d)None of these
11. Which of the following is definitely true ?
    (a)C is third to the right of D
    (b)D is facing P
    (c)C is facing S
    (d)None of these
12. Who is standing to the immediate right of A?
    (a)E (b)C (c)D (d)B
13. Who is facing B?
    (a)R (b)S (c)Q (d)T
14. The sum of ages of a daughter and mother is 63 yr. Four years back mother's age was 4 times that of daughter's age at that time. What is present age of mother?
    (a)46 yr (b)48 yr
    (c)50 yr (d)59 yr
15. A watch gains 10 s in 5 min was set coreect at 9:00 am am. When the watch indicated 20 min past 7:00 pm in the same eveing, the correct time is
    (a)7:00 pm (b)7:40 pm
    (c)7:10 pm (d)8:00 pm
16. Father is aged three times more than the age of his son Rohit. After 8 yr, he would be two and a half times of Rohit's age. After further 8 yr, how many times would he be of Rohit's age?
    (a)2 times (b)3 times
    (c)2.5 times (d)3.5 times
17. What is the number that comes next in the series?
    1, 2, 3, 6, 11, 20, 37, 68,
    (a)105 (b)124 (c)125 (d)126

Direction[Q. Nos. 18 & 19]: Six friends A, B, C, D, E and F are sitting round a hexagonal table. F, who is sitting exactly opposite A, is to be immediate right of B, D is between A and B and is exactly opposite to C.

18. Who are stting next to A?
    (a)D and E (b)D and F
    (c)C and E (d)B and D
19. Who is sitting opposite to B?
    (a)A (b)C (c)E (d)F
20. The arithmetic mean of 2^{10} and 2^{20} is
    (a)2^{15} (b)$2^5 + 2^{10}$ (c)$2^9 + 2^{20}$ (d)$2^9 + 2^{19}$
21. There are five different boxes of different unknown weights each less than 100 kg. These boxes were weighted in pairs and the weights obtained are 110, 112, 113, 114, 115, 116, 117, 118, 120 and 121 kg. What is the weighted in kg of the heaviest box ?
    (a)60 (b)62 (c)64 (d)61

Direction[Q. Nos. 22 to 26] : All the roads of a city are either perpendicular or parallel to one another. The roads are all straight. Roads A, B, C, D and E are parallel to one another. Roads F, G, H, I, J, K, L and M are parallel to one another.
Road A is 1 km East of road B.
Road B is ½ km West of road C.
Road D is 1 km West of road E.
Road G is ½ km South of road H.
Road I is 1 km North of road J.
Road K is ½ km North of road L.
Road K is 1 km South of road M.

22. Which of the following is necessarily true?
    (a) E and B intersect
    (b) D is 2 km West Of B
    (c) D is atleast 2 km West of A
    (d) M is 1.5 km North of L
23. If E is between B and C, then which of the following is false?
    (a) D is 2 km West of A
    (b) C is less than 1.5 km from D
    (c) Distance from E to B added to distance of E to C is ½ km
    (d) E is less than 1 km from A
24. If road E is between B and C, then the distance between A and D is
    (a) less than 1 km
    (b) C is 1 km West of D
    (c) between ½ km and 2 km
    (d) more than 2 km
25. Which of the following possiblilities would make two roads coincide?
    (a) L is ½ km North of I   (b) C is 1 km West of D
    (c) I is ½ km North of K   (d) E and B are ½ km apart
26. If K is parallel to I, K is ½ km South of J and1 km North of G, then which of the following two roads would be ½ km apart?
    (a)I and K   (b)J and G
    (c)I and G   (d)J and K
27. The students in three classes are in the ratio 2:3:5. If 20 students are increased in each class, the ratio changes to 4:5:7. The total number of students before the increase were
    (a)10   (b)90
    (c)100   (d)None of these
28. Ajith is three times older than Babita. Chetu is half the age of Das. Babita is older than Chetu. Which of the following additional information is needed to estimate the age of Ajith?
    I. Chetu is 10 yr old.
    II. Both Babita and Das are older than Chetu by the same number of years.
    (a) Only I   (b) Only II
    (c) I and II   (d) None of these

Direction [Q. Nos. 29 to 32] : Six friend P, Q, R, S, T and U are standing in two rows facing one another. P is the middle of one row. U is to the left to S and facing R, Q and T are not in the same row. Only one person is in between R and T.

29. Which of the following are in the same row?
    (a)U, S and T   (b)R, P and T
    (c)U, Q and P   (d)U, R and Q
30. Who is to the left of S?
    (a)P   (b)U
    (c)S   (d)Q
31. Who faces P?
    (a)Q   (b)T
    (c)S   (d)U
32. Which of the following pairs are facing each other?
    (a)RS   (b)TU
    (c)PU   (d)TQ

Direction[Q. Nos. 33 to 37] : Six members of a family A, B, C, D, E and F are Psychologist, Manager, Advocate, Jeweller, Doctor and Engineer but not necessarily in same order. Doctor is the grandfather of F, who is Psychologist. Manager D is married to A.
Jeweller C is married to Advocate.
B is the mother of F and E.
There are two married couples in the family.

33. Which is the profession of A?
    (a)Manager   (b)Engineer
    (c)Can't be determined   (d)None of these
34. What is the profession of E?
    (a)Manager   (b)Engineer
    (c)Doctor   (d)None of these
35. How is A related to E?
    (a)Grandmother   (b)Wife
    (c)Grandfather   (d)None of these
36. How many male members are there in the family?
    (a)Two   (b)Three
    (c)Four   (d)Can't be determined
37. Who are the two couples in the family?
    (a)AD and CB   (b)AB and CD
    (c)AC and BD   (d)None of these

Direction[Q. Nos. 38 to 40] : At a small company, parking spaces are reserved for the top executives: CEO, President Vice-President, Secretary and Treasurer with the spaces lined up in that order. The parking lot guard can tell at a glance, if the cars are parked correctly by looking at the colour of the cars. The cars are yellow, green, purple, red and blue and the executive names are Alice, Bert, Cheryl, David and Enid.
The car in the first space is red.
A blue car is parked between the red car and the green car.
The car in the last space is purple.
The secretary drives a yellow car.
Alice's is parked next to David's.

Enid drives a green car.
Bert's car is parked between Cheryl's and Enid's.
David's car is parked in the last space.

38. Who is the secretary ?
(a)Enid (b)David
(c)Cheryl (d)Alice

39. Who is the CEO?
(a)Alice (b)Bert
(c)Cheryl (d)David

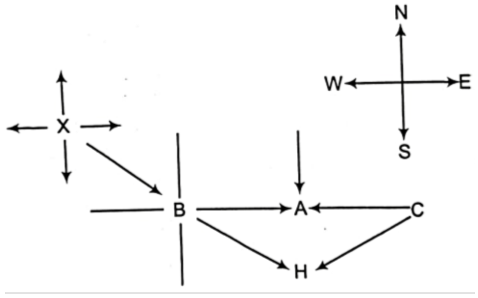
40. What colour is the Vice-President car?
(a)Green (b)Yellow
(c)Blue (d)Purple

ANSWER

1.	B	2.	C	3.	B	4.	D	5.	C
6.	D	7.	A	8.	B	9.	B	10.	A
11.	D	12.	A	13.	C	14.	B	15.	A
16.	A	17.	C	18.	A	19.	C	20.	D
21.	B	22.	D	23.	A	24.	C	25.	C
26.	D	27.	C	28.	C	29.	B	30.	B
31.	C	32.	D	33.	D	34.	B	35.	C
36.	D	37.	A	38.	D	39.	C	40.	a

SOLUTION

1. **(b)** On the basis of the given information regarding directions, we have the following figure.



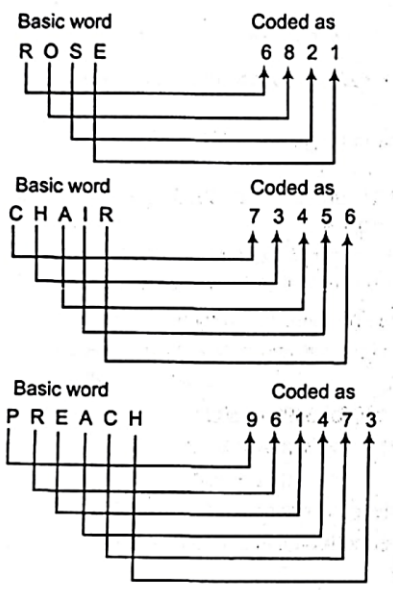
It is clear from the figure that H is at the farthest South and C is at the farthest East.
So, option (b) is correct.

2. **(c)** It is very clear from the given information that C does not work with A, therefore should be the partner with B as they can form pairs i.e. AD and CB.

So, according to the question,
Number of possibility for each player is

A	B	C	D	
A	×	×	×	✓
B	×	×	✓	×
C	×	×	×	✓
D	✓	×	×	×

3. **(b)**



Therefore, the coding for word SEARCH will be 214673.
Hence, option (b) is the correct choice.

Solutions [Q. Nos. 4 to 6]

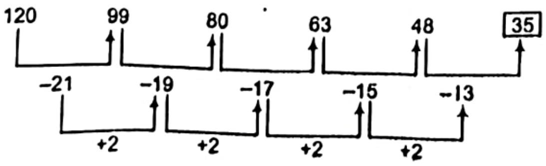
A table is drawn on the basis of the given information.

Town	A	B	C	D	E
Team	T	R	P	Q	S

4. (d) Team Q is from town D, according to the above table.
5. (c) Team P is from the town C, according to the above table.
6. (d) Team T is from the town A, according to the above table.
7. (a) Given series

   120, 99, 80, 63, 48, ...

   We have the following pattern



   Hence, the number that comes next in the series is 35.

8. (b) Total number of students when there were 24 students in each of

   13 sections = 13 × 24 = 312

   Total number of students when three new section were added with 21 students in each of 16 sections

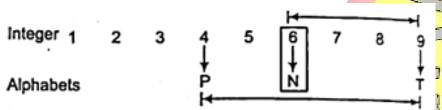
   = 21 × 16 = 336

   Number of newly admitted students = 336 – 312 = 24

9. (b) On the basis of the given information, the integer assigned to N is 6.

   This is explained below

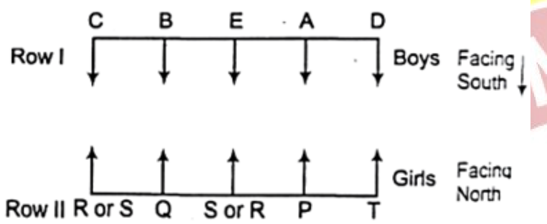
   The difference between N and Tis 3



   The difference between P and T is 5

Solutions [Q. Nos. 10 to 13]

On the basis of the given information, we have the standing arrangement as shown below.



10. (a) C and D are standing at the ends of the row.
11. (d) None of the option is correct.
12. (a) E is standing to the immediate right of A.
13. (c) Q is facing B.
14. (b) Let the present age of daughter = x yr

    and present age of mother = y yr

    According to the question,

    $$x + y = 63$$

    4yr back,

    Daughter's age = (x - 4) yr

    Mother's age = (y - 4) yr

    $\therefore$ $4(x – 4) = y – 4$

    $4x – 16 = y – 4$

    $4x – y = 12$

    From Eqs. (i) and (ii), we get

    $x + y = 63$

    $\Rightarrow$ $4x – y = 12$

    $\therefore$ $x = 15$ yr

    $\because$ $x + y = 63$

    $\Rightarrow$ $15 + y = 63$

    $\Rightarrow$ $y = 48$ yr

    Hence, present age of mother = 48 yr.

15. (a) Time gained in 5 min = 10 s

    So, time gained in 60 min = $\frac{10}{5} \times 60$

    = 120 s = 2 min

    So, time gained in 1 h = 2 min

    Time gained in 10 h = 2 × 10 min = 20 min

    Now,

    Present time = 7:20 pm

    So, correct time is 7 pm.

16. (a) Let the present age of Rohit = x yr

    and present age of Rohit 's father = x + 3x = 4 x yr

    After 8 yr,

    Rohit's age = x + 8 yr

    Rohit's father's age = 4x + 8 yr

    According to the question,

    $\Rightarrow$ $\frac{5}{2}(x + 8) = 4x + 8$

    $\Rightarrow$ $\frac{5x}{2} + 20 = 4x + 8$

    $\therefore$ $x = 8$

    i.e. Rohit age after 8 yr = 16 yr

    Rohit's father age after 8 yr = 40 yr

    Now, after further 8 yr

    Rohit age 16 + 8 = 24 yr

    Rohit 's father = 40 + 8 = 48 yr

    Hence, Rohit 's father age will be two times of Rohit age.

17. (c) 1, 2, 3, 6, 11, 20, 37, 68,..

    The above series is showing the following pattern. We have to the sum the three numbers

    i.e. 1 + 2 + 3 = 6

    2 + 3 + 6 = 11

    3 + 6 + 11 = 20

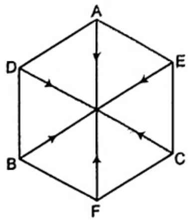
    11 + 20 + 37 = 68

    20 + 37 + 68 = 125

    So, the number that comes next in the series, is 125.

Solutions (Q. Nos. 18 to 19]

On the basis of the given information, sitting arrangement is shown below:



18. (a) D and E are sitting next to A.

19. (c) E is sitting opposite to B.

20. (d) Arithmetic mean of 2^{10} and 2^{20} = $\frac{2^{10}+2^{20}}{2}$

$= \frac{2^9+2^{19}}{2} = 2^9 + 2^{19}$

21. (b) Let the weights of the five boxes, in increasing order be A, B, C, D and E.

i.e. $A < B < C < D < E$.

Each of the boxes can be paired up with another box in a total of 4 ways.

If we sum up all the given weights, we get

$4(A + B + C + D + E) = (110 + 112 + 113 + 114 + 115 + 116 + 117 + 118 + 120 + 121)$

$= 1156$ kg

$\Rightarrow A + B + C + D + E = 289$ kg ...(i)

Now, 121 kg is the sum of the weight of the boxes D and E and 110 is the sum of the weights of the boxes A and B.

i.e. $A + B = 110$ kg

and $D + E = 121$kg

$\therefore A + B + D + E = 231$kg ...(ii)

Subtracting Eq. (ii) from Eq. (i), we get

$C = 289 - 231 = 58$ kg

Now, 120 kg is the sum of the weights of the boxes C and E

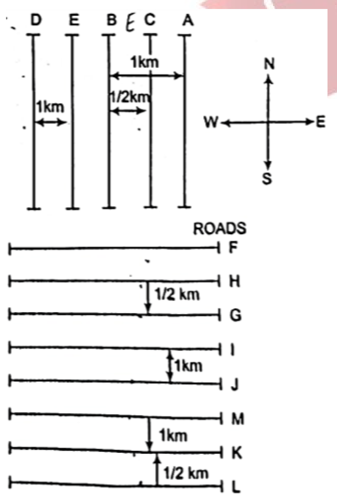
i.e. $C + E = 120$ kg

$E = (120 - 58)$ kg $= 62$ kg

Solutions [Q. Nos. 22 to 26]

On the basis of the given information in the question, we have the figure as shown below.

ROADS→



22. (d) M is 1.5 km North of L.

23. (a) "D is 2 km West of A" is not true.

24. (c) The distance between A and D lies between $\frac{1}{2}$ and 2 km.

25. (c) If I is 1/2 km North of (b), there is possibility of coinciding of the roads J and L.

26. (d) J and K would be 1/2 km apart.

27. (c) Let the students in three classes be 2x, 3x and 5x.

On increasing the students in each classes by 20, the students will be 2x + 20, 3x + 20 and 5x + 20. Now, according to the question,

$2x + 20 : 3x + 20 : 5x + 20 :: 4 : 5 : 7$

$\because \quad \frac{2x+20}{3x+20} = \frac{4}{5}$

$10x + 100 = 12x + 80$

$x = 10$

Total number of students before increase were

$= 2x + 3x + 5x = 10x = 10(10) = 100$

28. (c) Let the ages of Ajith, Babita, Chetu, Das are A, B, C, D yr. Age of Ajith is three times that of Babita i.e. A = 3B

Chetu is half the age of Das i.e. $C = \frac{D}{2}$

Babita is older than Chetu i.e. B > C.

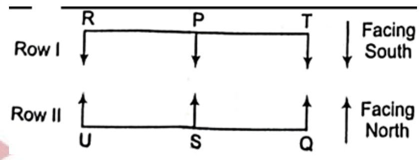
Statement I Chetu is 10 yr old. This statement is necessary to find the age of Ajith.

Statement II This statement is again necessary to find the age of Ajith.

Hence, both statements I and II are necessary to estimate the age of Ajith.

Solutions [Q. Nos. 29 to 32]

On the basis of the information given for the questions, we have the following standing arrangement of six friends i.e. P, Q, R, S, T, U.



29. (b) R, P and T are in the same row.

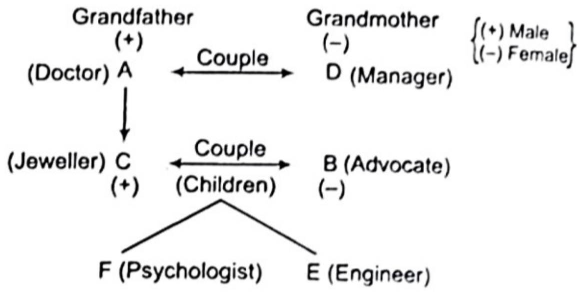
30. (b) U is standing to the left of S.

31. (c) S is facing P.

32. (d) T is standing opposite to Q, i.e. TQ are facing each other:

Solutions [Q. Nos. 33 to 37]

On the basis of the given information, we have the following 1 relationship diagram.



33. (d) Profession of A is doctor. But it is not given in any of the options.
34. (b) Profession of E is engineer.
35. (c) A is grandfather of E.
36. (d) This cannot be determined.
37. (a) AD and CB are the two couples in the family.

Solutions [Q. Nos. 38 to 40]

Based on the given information, following table is drawn.

Executive	CEO	President	Vice – President	Secretary	Treasurer
Executive names	Cheryl	Bert	Enid	Alice	David
Cars	Red	Blue	Green	Yellow	Purple
Order	1	2	3	4	5

38. (d) Alice is the secretary.
39. (c) Cheryl is the CEO.
40. (a) Vice-President's car is of green colour.